



11420 - On the shoulders of a giant: towards a chemical inventory of Herbig discs

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Alice Somigliana (PI) (ESA Member)	Max Planck Institute for Astronomy
Dr. Giulia Perotti (CoI) (ESA Member)	Max Planck Institute for Astronomy
Ms. Jenny Frediani (CoI) (ESA Member)	Stockholm University
Dr. Haochang Jiang (CoI) (ESA Member)	Max Planck Institute for Astronomy
Dr. Lucas Stapper (CoI) (ESA Member)	Max Planck Institute for Astronomy
Dr. Myriam Benisty (CoI) (ESA Member)	Max Planck Institute for Astronomy
Dr. Arjan Bik (CoI) (ESA Member)	Stockholm University

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	IRAS23077 bkg SHOR T	MIRI Medium Resolution Spectroscopy	(2) IRAS23077+6707-BKG
	2	IRAS23077 SHORT	MIRI Medium Resolution Spectroscopy	(1) IRAS23077+6707
	3	IRAS23077 bkg MEDI UM	MIRI Medium Resolution Spectroscopy	(2) IRAS23077+6707-BKG
	4	IRAS23077 MEDIUM	MIRI Medium Resolution Spectroscopy	(1) IRAS23077+6707
	5	IRAS23077 bkg LONG	MIRI Medium Resolution Spectroscopy	(2) IRAS23077+6707-BKG
	6	IRAS23077 LONG	MIRI Medium Resolution Spectroscopy	(1) IRAS23077+6707

ABSTRACT

The chemical composition of protoplanetary discs strongly impacts the planet formation process, especially through the molecular ices coating dust grains, which directly build up planetary cores. In Class II objects, the absorption features of ice species are only visible in the edge-on configuration,

where the light coming from the central star and the warm inner disc is blocked. With the unprecedented sensitivity and spatial resolution of JWST/MIRI-MRS, we now have spectra of edge-on, Class II sources, but the almost totality of the available data are limited to discs around low mass, late spectral type stars. The most typical formation site of giant planets, Herbig discs (around massive, earlier spectral type stars), pose an additional challenge: their high stellar luminosities, and bright continuum emission from the inner disc, readily saturate the detectors. We propose to obtain MIRI-MRS observations of the recently discovered, edge-on Herbig disc IRAS 23077+6707. Its isolated location is ideal to understand the impact of the stellar UV field on the disc chemistry, without the contribution of external irradiation; furthermore, its edge-on configuration allows to both detect ice absorption features and observe the target without reaching saturation. We will obtain a detailed chemical inventory, mapping the radial and vertical chemical gradient across the disc; furthermore, we will constrain the presence of PAHs, as well as the thermal processing of dust grains. Comparing these results with the spectra of lower mass stars will unveil the impact of stellar radiation on ice chemistry, and redefine our understanding of the environment where massive planets form.

OBSERVING DESCRIPTION

We request $S/N > 500$ up to channel 3 long, where the ices and most of the PAH features are, and > 200 in channel 4. This S/N is enough to detect faint ice bands. We adjusted the number of groups for this purpose and adopted a 4 dither pattern in extended source mode. To cover the whole extent of our source, we set up an 8-tile mosaic with the recommended 10 % overlap, arranged as a 4x2 pattern. We apply a row and column shift of -20 and -40 respectively, to ensure overlap with the fangs in the northern part of the disc. To compensate the difficulty in obtaining a proper background subtraction with such an extended source, we request a single tile background observation to each visit, chosen as an emission-free region close to DraChi inspecting the surrounding area with Aladin. The background observations are linked in a non-interruptible sequence to the science observations.

Proposal 11420 - Targets - On the shoulders of a giant: towards a chemical inventory of Herbig discs

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	IRAS23077+6707	RA: 23 09 43.6450 (347.4318542d) Dec: +67 23 38.94 (67.39415d) Equinox: J2000	Epoch of Position: 2015.5	
	Comments: Category=Star Description=[Protoplanetary disks] Extended=YES				
Fixed Targets	(2)	IRAS23077+6707-BKG	RA: 23 09 44.1362 (347.4339008d) Dec: +67 23 21.71 (67.38936d) Equinox: J2000		
	Comments: Category=Star Description=[Protoplanetary disks]				

Proposal 11420 - Observation 1 - On the shoulders of a giant: towards a chemical inventory of Herbig discs

Observation	Proposal 11420, Observation 1: IRAS23077 bkg SHORT												Fri Mar 13 21:05:38 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [IRAS23077 SHORT (Obs 2)]												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	IRAS23077+6707-BKG	RA: 23 09 44.1362 (347.4339008d) Dec: +67 23 21.71 (67.38936d) Equinox: J2000										
	Comments: Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
	F560W	All MRS			NO			FULL			Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				BACKGROUND				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	27	4	1	Dither 1	2	8	616.059	
	1	SHORT(A)	MRSSHORT		FASTR1	27	4	1	Dither 1	2	8	616.059	

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 11420 - Observation 2 - On the shoulders of a giant: towards a chemical inventory of Herbig discs

Observation	Proposal 11420, Observation 2: IRAS23077 SHORT													Fri Mar 13 21:05:38 GMT 2026
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[IRAS23077 bkg SHORT (Obs 1)]													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	IRAS23077+6707			RA: 23 09 43.6450 (347.4318542d) Dec: +67 23 38.94 (67.39415d) Equinox: J2000			Epoch of Position: 2015.5						
	Comments: Category=Star Description=[Protoplanetary disks] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	FND	All MRS			NO			FULL			Allow Auto Reorder			
Mosaic	Rows	Columns		Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	4	2		10.0		10.0		-20.0		-40.0		DEFAULT		
Dithers	#	Dither Type					Optimized For				Direction			
	1	4-Point					EXTENDED SOURCE				NEGATIVE			
	2	2-Point					POINT SOURCE				POSITIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SHORT(A)	MRSLONG		FASTR1	27	2	1	Dither 1	4	8	610.509	267182	
	1	SHORT(A)	MRSSHORT		FASTR1	27	2	1	Dither 1	4	8	610.509	267182	
	2	SHORT(A)	MRSLONG		FASTR1	27	2	1	Dither 2	2	4	305.254	267182	
	2	SHORT(A)	MRSSHORT		FASTR1	27	2	1	Dither 2	2	4	305.254	267182	

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 11420 - Observation 3 - On the shoulders of a giant: towards a chemical inventory of Herbig discs

Observation	Proposal 11420, Observation 3: IRAS23077 bkg MEDIUM												Fri Mar 13 21:05:38 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [IRAS23077 MEDIUM (Obs 4)]												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	IRAS23077+6707-BKG		RA: 23 09 44.1362 (347.4339008d) Dec: +67 23 21.71 (67.38936d) Equinox: J2000									
	Comments: Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS			NO			FULL			Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				BACKGROUND				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM(B)	MRSLONG		FASTR1	27	4	1	Dither 1	2	8	616.059	
	1	MEDIUM(B)	MRSSHORT		FASTR1	27	4	1	Dither 1	2	8	616.059	

Special Requirements	Sequence Observations 3, 4, Non-interruptible
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Proposal 11420 - Observation 4 - On the shoulders of a giant: towards a chemical inventory of Herbig discs

Observation	Proposal 11420, Observation 4: IRAS23077 MEDIUM												Fri Mar 13 21:05:38 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[IRAS23077 bkg MEDIUM (Obs 3)]												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	IRAS23077+6707	RA: 23 09 43.6450 (347.4318542d) Dec: +67 23 38.94 (67.39415d) Equinox: J2000				Epoch of Position: 2015.5						
	Comments: Category=Star Description=[Protoplanetary disks] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	FND	All MRS				NO			FULL		Allow Auto Reorder		
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	4	2	10.0		10.0		-20.0		-40.0		DEFAULT		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
	2	4-Point				POINT SOURCE				POSITIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM(B)	MRSLONG		FASTR1	27	2	1	Dither 1	4	8	610.509	267182
	1	MEDIUM(B)	MRSSHORT		FASTR1	27	2	1	Dither 1	4	8	610.509	267182
	2	MEDIUM(B)	MRSLONG		FASTR1	27	2	1	Dither 2	4	8	610.509	267182
	2	MEDIUM(B)	MRSSHORT		FASTR1	27	2	1	Dither 2	4	8	610.509	267182

Special Requirements	Sequence Observations 3, 4, Non-interruptible
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Proposal 11420 - Observation 5 - On the shoulders of a giant: towards a chemical inventory of Herbig discs

Observation	Proposal 11420, Observation 5: IRAS23077 bkg LONG												Fri Mar 13 21:05:38 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [IRAS23077 LONG (Obs 6)]												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	IRAS23077+6707-BKG	RA: 23 09 44.1362 (347.4339008d) Dec: +67 23 21.71 (67.38936d) Equinox: J2000										
	Comments: Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				NO			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				BACKGROUND				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	LONG(C)	MRSLONG		FASTR1	27	4	1	Dither 1	2	8	616.059	
	1	LONG(C)	MRSSHORT		FASTR1	27	4	1	Dither 1	2	8	616.059	

Special Requirements	Sequence Observations 5, 6, Non-interruptible
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Proposal 11420 - Observation 6 - On the shoulders of a giant: towards a chemical inventory of Herbig discs

Observation	Proposal 11420, Observation 6: IRAS23077 LONG												Fri Mar 13 21:05:38 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[IRAS23077 bkg LONG (Obs 5)]													
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	IRAS23077+6707			RA: 23 09 43.6450 (347.4318542d) Dec: +67 23 38.94 (67.39415d) Equinox: J2000			Epoch of Position: 2015.5						
	Comments: Category=Star Description=[Protoplanetary disks] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	FND	All MRS			NO			FULL			Allow Auto Reorder			
Mosaic	Rows	Columns		Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	4	2		10.0		10.0		-20.0		-40.0		DEFAULT		
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					POINT SOURCE			NEGATIVE				
	2	4-Point					POINT SOURCE			POSITIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	LONG(C)	MRSLONG		FASTR1	27	2	1	Dither 1	4	8	610.509	267182	
	1	LONG(C)	MRSSHORT		FASTR1	27	2	1	Dither 1	4	8	610.509	267182	
	2	LONG(C)	MRSLONG		FASTR1	27	2	1	Dither 2	4	8	610.509	267182	
	2	LONG(C)	MRSSHORT		FASTR1	27	2	1	Dither 2	4	8	610.509	267182	

Special Requirements	Sequence Observations 5, 6, Non-interruptible
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